# Mathematical Formulation of the Proactive Golden-Hour Bridge Model (PGBM) for Post-Earthquake UAV-Assisted Response

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***Abstract*—This document presents the formal mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM), a UAV-assisted emergency response system for post-earthquake scenarios. The system operates as an online decision-making framework where Responder UAVs are dynamically assigned to detected survivors to provide immediate assistance during the critical "Golden Hour" before ground rescue teams physically arrive. The model captures the coupling between survivor task states, UAV dynamics, battery constraints, time-dependent urgency decay, and 3D routing feasibility. The global system state is formally defined, and the event-triggered decision policy is mathematically characterized.**

## I. Introduction

In the aftermath of a major earthquake, Scout UAVs can rapidly detect trapped survivors using thermal and optical sensors. However, Ground Rescue Teams face significant delays due to collapsed infrastructure and blocked roads, creating a deadly gap between detection and physical rescue. The Proactive Golden-Hour Bridge Model (PGBM) introduces an intermediate UAV-assisted response layer where Responder UAVs are dispatched immediately to provide emergency supplies, two-way communication, and GPS relay before ground teams arrive.

### A. System Overview

The PGBM consists of four primary actors:

1) **Scout UAVs**: Detect survivors and transmit detection packets.
2) **Command Center**: Maintains global state and executes online allocation/routing decisions.
3) **Responder UAVs**: Execute bridge missions to deliver aid.
4) **Ground Rescue Team**: Performs final physical extraction (outside core model).

### B. Core Assumptions

- Obstacles are static during a single mission (Version 1).
- Communication between all actors is reliable.
- Each Responder UAV executes one tour at a time, serving up to $K$ demand items (possibly multiple survivors, e.g., $K = 4$).
- Survivor tasks are independent except via shared tour capacity $K$, inventory $I_j$, and tour energy $E(\pi_j)$; each tour is non-preemptive (once dispatched, the order is fixed until return to base).

## II. Mathematical Notation

Throughout this formulation, bold uppercase letters denote sets, bold lowercase denote vectors, and scalar values use standard notation. The time variable $t$ denotes the current decision time (continuous). All decisions are event-triggered.

TABLE I
Notation Summary

| Symbol | Description |
|---|---|
| $\mathcal{S}(t), \mathcal{D}(t), \mathcal{E}$ | Survivor set, Responder UAV set, Environment |
| $\mathcal{Q}_{\text{high}}, \mathcal{Q}_{\text{low}}$ | High/low confidence queues ($c_i \geq c_{\text{thresh}}$) |
| $S_i = (p_i, t_i^{\text{det}}, c_i, \sigma_i, \mathcal{R}_i, a_i, \text{assigned\_to}_i)$ | Survivor task with demand set $\mathcal{R}_i$ |
| $D_j = (p_j, p_j^{\text{base}}, B_j, K, I_j, m, A_j, s_j, \text{status}_j)$ | Responder UAV with capacity $K$, inventory $I_j$, payload $m$ |
| $p_i, p_j, p_j^{\text{base}} \in \mathbb{R}^3$ | Positions and base station |
| $\sigma_i, c_i \in [0,1], \mathcal{R}_i$ | Severity, confidence, demand set $\subseteq \{\text{med}, \text{water}, \text{food}\}$ |
| $u_i(t) = \sigma_i e^{-\alpha(t - t_i^{\text{det}})}$ | Derived urgency, $\alpha = \ln 2 / T_{\text{half}}$ |
| $B_j(t) \in [0, B_{\max}], B_{\max}, E_{\text{reserve}}$ | Battery, max, reserve $0.2 B_{\max}$ |
| $A_j \in \{0,1\}, \text{status}_j, s_j$ | Availability (Idle), status, current tour $\pi_j$ |
| $T(\pi_j), E(\pi_j), C(\pi_j)$ | Tour time, energy, cost (primary); $T_{ij}, E_{ij}, C_{ij}$ is $k=1$ special case |
| $T_{\max}, E_{\max}, c_{\text{thresh}}, \alpha, T_{\text{half}}, K$ | Normalization, threshold, half-life, capacity |
| $\pi_j, P_{\pi_j}, x_{ij}$ | Tour $(i_1, \ldots, i_k)$, tour route, single-leg assignment ($k=1$) |
| $X(t) = (\mathcal{S}, \mathcal{D}, \mathcal{E}, \mathcal{Q})$ | Global state ($\mathcal{Q} = \mathcal{Q}_{\text{high}} \cup \mathcal{Q}_{\text{low}}$) |

## III. System Entities and State Definitions

### A. Survivor Task Model

Each survivor detection is modeled as a task $i \in \mathcal{S}(t)$, where $\mathcal{S}(t)$ is the set of all known survivor tasks at time $t$. Survivor $i$ is defined as:

$$S_i = (x_i, y_i, z_i, t_i^{\text{det}}, c_i, \sigma_i, \mathcal{R}_i, a_i, \text{assigned\_to}_i) \tag{1}$$

with the following parameters:

TABLE II
Survivor Task Parameters

| Parameter | Symbol | Domain |
|---|---|---|
| Location | $(x_i, y_i, z_i)$ | $\mathbb{R}^3$ |
| Detection Time | $t_i^{\text{det}}$ | $\mathbb{R}^+$ |
| Confidence Score | $c_i$ | $[0, 1]$ |
| Severity | $\sigma_i$ | $[0, 1]$ |
| Demand Set | $\mathcal{R}_i$ | $\subseteq \{\text{med}, \text{water}, \text{food}\}$ |
| Assignment Flag | $a_i$ | $\{0, 1\}$ |
| Assigned UAV | $\text{assigned\_to}_i$ | $\mathcal{D}(t) \cup \{\emptyset\}$ |

$\mathcal{R}_i$ is the set of relief items required by $i$ (e.g., $\{\text{med}\}$, $\{\text{med}, \text{water}\}$) provided by the Scout/Command Center; $|\mathcal{R}_i| \geq 1$ and $|\mathcal{R}_i| > 1$ denotes multiple demands at the same location. Severity $\sigma_i$ denotes the static medical criticality at detection (independent of time), with $\sigma_i = 1$ representing the most critical condition. Two possible scenarios for obtaining $\sigma_i$ are: (1) *Scout AI Inference*: estimated automatically from Scout UAV thermal/optical imagery and rubble context via a classification model (e.g., motion/heat signature analysis), and (2) *Human-in-the-Loop*: tagged manually by the Command Center operator reviewing the Scout video feed.

### B. Urgency Decay Function

The urgency of survivor $i$ decays exponentially with waiting time according to the Golden Hour principle:

$$u_i(t) = \sigma_i \cdot e^{-\alpha(t - t_i^{\text{det}})} \tag{2}$$

where $\sigma_i \in [0, 1]$ is the severity at detection so that $u_i(t_i^{\text{det}}) = \sigma_i$, $\Delta t = t - t_i^{\text{det}} \geq 0$ is the waiting time, and $\alpha > 0$ is the decay rate. The rate is defined via the half-life $T_{\text{half}}$ (e.g., $T_{\text{half}} = 30$ min) as

$$\alpha = \frac{\ln 2}{T_{\text{half}}}, \tag{3}$$

so that $u_i(t_i^{\text{det}} + T_{\text{half}}) = \sigma_i/2$. For $T_{\text{half}} = 30$ min, $\alpha \approx 0.023$ min$^{-1}$ ($3.8 \times 10^{-4}$ s$^{-1}$); $T_{\text{half}} = 60$ min gives $\alpha \approx 0.0115$ min$^{-1}$. A survivor found 10 min ago ($\Delta t = 10$) has $u \approx 0.79\sigma_i$, while one found 90 min ago has $u \approx 0.12\sigma_i$ at the same current time $t$, capturing higher salvageability of fresher detections.

### C. Confidence Score and Verification Policy

Each detection carries a confidence score $c_i \in [0, 1]$ representing the Scout UAV detector's estimated probability that the detection is a true survivor (e.g., thermal/optical AI output). A threshold $c_{\text{thresh}} \in [0, 1]$ (e.g., $0.5$) separates high- and low-confidence regimes:

$$\text{eligible}(i) = \begin{cases} 1 & \text{if } c_i \geq c_{\text{thresh}} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

Detections with $c_i \geq c_{\text{thresh}}$ enter the high-confidence queue $\mathcal{Q}_{\text{high}}(t)$ and are eligible for Responder assignment. Detections with $c_i < c_{\text{thresh}}$ enter the low-confidence queue $\mathcal{Q}_{\text{low}}(t)$ and are *not* eligible for Responder dispatch; they require verification (Scout re-scan or operator review) before promotion to $\mathcal{Q}_{\text{high}}$. In Version 1, $\mathcal{Q}_{\text{low}}$ is modeled as a holding set and verification is abstracted as an exogenous promotion event. The offloading ILP is defined only over $\mathcal{Q}_{\text{high}}(t)$.

### D. Responder UAV Model

Each Responder UAV $j \in \mathcal{D}(t)$ is defined as:

$$D_j = (x_j, y_j, z_j, p_j^{\text{base}}, B_j, K, I_j, m, A_j, s_j, \text{status}_j) \tag{5}$$

with the following parameters:

TABLE III
Responder UAV Parameters

| Parameter | Symbol | Domain |
|---|---|---|
| Position | $(x_j, y_j, z_j)$ | $\mathbb{R}^3$ |
| Base Station | $p_j^{\text{base}}$ | $\mathbb{R}^3$ |
| Remaining Battery | $B_j$ | $\mathbb{R}^+$ |
| Capacity | $K$ | $\mathbb{N}$ (e.g., 4 items) |
| Inventory | $I_j$ | multiset $\subseteq \{\text{med}, \text{water}, \text{food}\}$ |
| Payload Mass | $m$ | $\mathbb{R}^+$ (kg, $m = \sum_{item \in I_j} weight(item)$) |
| Availability | $A_j$ | $\{0, 1\}$ |
| Current Task | $s_j$ | $\mathcal{S}(t) \cup \{\emptyset\}$ |
| Mission Status | $\text{status}_j$ | $\{\text{Idle}, \text{EnRoute}, \text{AtSurvivor}, \text{Returning}\}$ |

$K$ is the maximum number of items, $I_j$ is the loaded inventory (e.g., $[\text{med}, \text{water}, \text{food}, \text{med}]$), and $m$ is the derived total weight of $I_j$ which decreases after each drop and affects $e_{\text{travel}}$ per leg.

### E. Availability States

The availability $A_j$ is derived from the UAV's status:

$$A_j = \begin{cases} 1 & \text{if status}_j = \text{Idle} \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

### F. Environment Model

The disaster environment is represented as continuous sets:

$$\mathcal{E} = (V, O) \tag{7}$$

where $V \subset \mathbb{R}^3$ is navigable free space and $O \subset \mathbb{R}^3$ is the static obstacle set (buildings, rubble, no-fly zones), with $V \cap O = \emptyset$. For Version 1, $O(t) = O$ (static) and the hazard map $R : V \to [0, 1]$ is set to $R = 0$ and deferred. A route $P$ is feasible iff $P \subset V$ (all waypoints lie in free space).

### G. Task Queue

The ordered queue of unassigned survivors is defined as a derived view:

$$\mathcal{Q}(t) = \text{order}(\{S_i \in \mathcal{S}(t) \mid a_i = 0\}) \tag{8}$$

ordered by priority:

$$\text{priority}(i) = \beta_1 \cdot u_i(t) + \beta_2 \cdot c_i \tag{9}$$

Note: $\mathcal{Q}(t)$ is not an independent state component but a derived view of $\mathcal{S}(t)$ for scheduling. With verification threshold $c_{\text{thresh}}$, the queue partitions as:

$$\mathcal{Q}_{\text{high}}(t) = \{S_i \in \mathcal{Q}(t) \mid c_i \geq c_{\text{thresh}}\}, \tag{10}$$

$$\mathcal{Q}_{\text{low}}(t) = \{S_i \in \mathcal{Q}(t) \mid c_i < c_{\text{thresh}}\} \tag{11}$$

Only $\mathcal{Q}_{\text{high}}(t)$ is eligible for the offloading assignment; $\mathcal{Q}_{\text{low}}(t)$ is held for verification. *Uncertainty handling:* The partition manages detection uncertainty (false positives from Scout thermal/optical AI). High-confidence detections are directly eligible, while low-confidence detections are buffered in $\mathcal{Q}_{\text{low}}$ for verification (re-scan/operator review) before promotion, preventing scarce Responder UAVs from being wasted on likely false alarms. Note: priority$(i)$ is a pre-filter ordering heuristic for $\mathcal{Q}_{\text{high}}(t)$ (e.g., for greedy fallback or tie-breaking); the optimal assignment is determined by the ILP cost $C_{ij}(t)$, not by priority order.

## IV. Decision Variables

### A. Tour Variable (Primary Decision)

At any decision time $t$, the offloading decision for each available UAV $j$ is an ordered tour

$$\pi_j(t) = (i_1, i_2, \ldots, i_k), \quad k = \text{len}(\pi_j), \quad \sum_{i \in \pi_j} |\mathcal{R}_i| \leq K \tag{12}$$

where $i_l \in \mathcal{Q}_{\text{high}}(t)$ is the $l$-th survivor to be served, $|\mathcal{R}_i|$ is the number of items demanded by $i$, and $K$ (e.g., 4) is the UAV item capacity. Thus $\pi_j$ encodes both *selection* (which survivors) and *order* (in which sequence). For example, $\pi_j = (A, C)$ with $\mathcal{R}_A = \{\text{med}\}$ and $\mathcal{R}_C = \{\text{water}, \text{food}\}$ carries $|\pi_j| = 3 \leq 4$ items to two survivors; $\pi_j = (C, A)$ is a different tour with different travel distance, climb, and payload-dependent energy. An empty tour $\pi_j = \emptyset$ means the UAV stays idle at base. The set of all tours $\{\pi_j\}_{j \in \mathcal{D}_{\text{avail}}}$ partitions $\mathcal{Q}_{\text{high}}$ (each survivor in at most one tour).

### B. Tour Route (Geometric Realization)

A tour $\pi_j$ is realized geometrically as the concatenation of its $k + 1$ optimal leg paths:

$$P_{\pi_j} = P^*_{p_j \to p_{i_1}} \circ P^*_{p_{i_1} \to p_{i_2}} \circ \cdots \circ P^*_{p_{i_k} \to p_j^{\text{base}}} \tag{13}$$

where each leg $P^*_{a \to b} = \{a = p_0, p_1, \ldots, p_L = b\}$ is the cheapest feasible waypoint sequence with $p_k \in V$ (e.g., $P^*_{p_j \to i_1} = \{p_j, x_1, x_2, p_{i_1}\}$ avoiding $O$). If any leg has no feasible path in $V$, the whole tour is infeasible. The routing oracle provides each leg: RoutingOracle$(p_a, p_b) \to (T_{ab}, E_{ab})$ or $(\infty, \infty)$ if infeasible. The tour aggregates $T(\pi_j) = \sum_l T_l + k \cdot T_{\text{hover}}$ and $E(\pi_j) = \sum_l E_l(m_{l-1}) + \sum_{i \in \pi_j} E_{\text{hover}}$, where $m_{l-1}$ is the payload before leg $l$ and thus early legs cost more. Offloading optimizes $\min_{\{\pi_j\},\text{order}} \sum_j C(\pi_j)$, i.e., both selection and order (TSP-like; $k!$ orders per set evaluated via the oracle).

## V. Cost Functions

### A. Path Cost

The cost of a path $P = \{p_0, \ldots, p_L\}$ is defined as its energy:

$$\text{Cost}(P) = E(P) = \sum_{k=1}^{L} (e_{\text{travel}}(p_k, m, w_0) \|p_k - p_{k-1}\| + e_{\text{vert}}(m) \max(0, \tag{14}$$

where $L$ is the number of waypoints and $e_{\text{travel}}, e_{\text{vert}}$ are as defined below. Thus $w_d, w_z$ are absorbed into $e_{\text{travel}}, e_{\text{vert}}$ and $\text{Cost}(P) = E(P)$.

### B. Optimal Path

The cheapest feasible path from $p_j$ to $p_i$ is

$$P^*_{ij} = \arg \min_{P \subset V,\, p_0 = p_j,\, p_L = p_i} \text{Cost}(P) \tag{15}$$

where the minimum is taken over all paths lying entirely in $V$. If no such path exists, $P^*_{ij}$ is infeasible and the pair $(i, j)$ is unreachable. When feasible, the travel time $T_{ij}$ and energy $E_{ij}$ are both derived from this single path $P^*_{ij}$. The routing oracle encapsulates this: given any two points it returns the corresponding $T$, $E$, or infinity if no feasible path exists. Offloading queries the oracle twice, for the outbound leg $p_j \to p_i$ and the return leg $p_i \to p_j^{\text{base}}$; both must be feasible for the assignment to be feasible.

### C. Travel Time per Tour

For a tour $\pi_j = (i_1, \ldots, i_k)$, the total travel time is the sum over legs plus per-stop hover:

$$T(\pi_j) = \sum_{l=0}^{k} T_{\text{leg}}(l) + \sum_{i \in \pi_j} T_{\text{hover}}(i), \quad T_{\text{leg}}(l) = \frac{\|P_l\|}{v_{\max}} \tag{16}$$

where $P_0 = P^*_{p_j \to p_{i_1}}$, $P_l = P^*_{p_{i_l} \to p_{i_{l+1}}}$ for $1 \le l < k$, $P_k = P^*_{p_{i_k} \to p_j^{\text{base}}}$, $v_{\max}$ is the maximum speed, and $T_{\text{hover}}(i) = T_{\text{drop}} + T_{\text{verify}} = 10\text{s} + 5\text{s} = 15$ s per stop (V1 fixed for drop and confirmation; future $\sigma$-dependent variable).

### D. Energy per Tour

Let $m_0 = \sum_{item \in I_j} weight(item)$ be the initial payload and $m_l$ the remaining payload before leg $l$ ($m_l = m_{l-1} - \sum_{item \in \mathcal{R}_{i_l}} weight(item)$). The energy of leg $l$ is

$$E_{\text{leg}}(l) = e_{\text{travel}}(p_l, m_{l-1}, w_0)\|P_l\| + e_{\text{vert}}(m_{l-1})\Delta z_l \quad (17)$$

with $e_{\text{travel}}(p, m, w_0) = e_0 + k_m m + k_w w_0$ and $e_{\text{vert}}(m) = mg/\eta_{\text{vert}}$ as above ($m, w_0$ realtime; $w_0$ is uniform wind scalar, $w(p)$ deferred). The total tour energy is

$$E(\pi_j) = \sum_{l=0}^{k} E_{\text{leg}}(l) + \sum_{i \in \pi_j} E_{\text{hover}}(i), \quad E_{\text{hover}}(i) = e_{\text{hover}} T_{\text{hover}}(i) \quad (18)$$

with $T_{\text{hover}}(i) = 15$ s (10s drop + 5s verify) per stop. Early legs use larger $m$ and thus higher $e_{\text{travel}}$.

### E. Total Energy per Tour

The total energy for tour $\pi_j$ is $E(\pi_j)$ as defined above; the single-leg $E_{ij}^{\text{total}}$ is the special case $k = 1$. where $P^*_{ij} = P^*_{j \to i}$. Variant $E_{ij}^{\text{oneway}} = E(P^*_{ij}) + E_{\text{hover}}$ (landing) is deferred; V1 assumes roundtrip.

### F. Coupling to Offloading

Routing provides $(T(\pi_j), E(\pi_j))$ via the oracle for each tour. Offloading uses these values in the feasibility check (20) and the tour cost (19). All $k+1$ legs $p_j \to p_{i_1} \to \cdots \to p_{i_k} \to p_j^{\text{base}}$ must be feasible; otherwise $C(\pi_j) = \infty$.

$$\mathcal{Q}_{\text{high}}(t),\ \mathcal{D}_{\text{avail}}(t)$$
$$\downarrow$$
$$\text{For each tour } \pi_j = (i_1, \dots, i_k)$$
$$\downarrow$$
$$\text{For each leg } l \text{ query Oracle}(p_l, p_{l+1}) \to (T_l, E_l) \text{ or } \infty$$
$$\downarrow$$
$$\text{Feasible}(\pi_j)? \quad E(\pi_j) + E_{\text{res}} \le B_j \wedge \mathcal{R}_{i_l} \subseteq I_j^{(l-1)} \wedge P_l \subset V$$
$$\downarrow$$
$$C(\pi_j) = \textstyle\sum_{i \in \pi_j} [w_T \frac{T}{T_{\max}} + w_E \frac{E}{E_{\max}} - w_U u_i c_i] \text{ or } \infty$$
$$\downarrow$$
$$\min_{\{\pi_j\}} \sum_j C(\pi_j) \ \to\ \{\pi_j^*\}$$
$$\downarrow$$
$$\text{Dispatch } \{\pi_j^*\}, \text{ update } X(t^+)$$

Fig. 1. PGBM coupling flow: Offloading queries Routing oracle per leg, aggregates $T(\pi_j), E(\pi_j)$, checks feasibility, computes $C(\pi_j)$, solves tour ILP, and updates state.

### G. Assignment Cost

The cost of assigning UAV $j$ to survivor $i$ is:

$$C(\pi_j, t) = \begin{cases} \sum_{i \in \pi_j} \left[ w_T \frac{T_{\text{leg}}(i)}{T_{\max}} + w_E \frac{E_{\text{leg}}(i)}{E_{\max}} - w_U u_i(t) c_i \right] & \text{if Feasible}(\pi_j \\ \infty & \text{otherwise} \end{cases} \quad (19)$$

where $w_T, w_E, w_U > 0$ are tour cost weights ($w_T + w_E + w_U = 1$ baseline), $T_{\max} = \max_{\text{feasible}} T_{\text{leg}}$ (e.g., 600 s for 2km map at $v_{\max} = 10$ m/s) and $E_{\max} = B_{\max}$ (e.g., 50000 J) are normalization constants (alternatively adaptive $T_{\max}(t) = \max_{i \in \mathcal{Q}_{\text{high}}} T_{ij}$ per snapshot), and $-w_U u_i c_i$ rewards urgency/confidence per survivor in the tour. Infeasible tours have $\infty$ cost. The single-pair $C_{ij}(t)$ is the special case $k = 1$ where $\pi_j = (i)$, i.e., $C_{ij} = C(\pi_j = (i))$.

*Rationale:* The tour cost is a dimensionless trade-off among time, energy, and expected value summed over the tour; $C_{ij}$ is $k = 1$.

## VI. Feasibility Constraints

### A. Battery Constraint for Tour

A tour $\pi_j$ is feasible only if the whole tour plus reserve fits the remaining battery:

$$E(\pi_j) + E_{\text{reserve}} \le B_j(t) \quad (20)$$

where $E(\pi_j) = \sum_l E_{\text{leg}}(l) + \sum_{i \in \pi_j} E_{\text{hover}}(i)$ from Step 4 and $E_{\text{reserve}} = 0.2\, B_{\max}$.

### B. Demand Matching

Each demand set must be satisfied from the remaining inventory:

$$\mathcal{R}_{i_l} \subseteq I_j^{(l-1)} \quad \forall i_l \in \pi_j \quad (21)$$

where $I_j^{(l-1)}$ is the inventory before stop $l$ ($I_j^{(0)} = I_j$). If $\mathcal{R}_{i_l} \not\subseteq I_j^{(l-1)}$, the tour is infeasible.

### C. Reachability per Leg

Each leg must lie in free space: $P_l \subset V$ for all $l$, i.e., $p_k \in V$ for every waypoint. Reachability is an oracle per leg; one infeasible leg makes the whole tour infeasible.

### D. Capacity and Availability

$\sum_{i \in \pi_j} |\mathcal{R}_i| \le K$ (total items) and $A_j = 1$ (Idle). If $A_j = 0$ or capacity exceeded, the tour is infeasible. Here $\text{len}(\pi_j) = k$ is the number of survivors and $\sum |\mathcal{R}_i|$ is the number of items.

### E. Feasible Tour Condition

A tour $\pi_j$ is feasible iff all of the above hold:

$$\begin{aligned}\text{Feasible}(\pi_j) =& (A_j = 1) \wedge (\sum_{i\in\pi_j} |\mathcal{R}_i| \le K) \wedge (\forall i \in \pi_j : c_i \ge c_{\text{thresh}}) \\ & \wedge (\forall l : P_l \subset V) \wedge (\forall i_l \in \pi_j : \mathcal{R}_{i_l} \subseteq I_j^{(l-1)}) \\ & \wedge (E(\pi_j) + E_{\text{reserve}} \le B_j(t))\end{aligned} \tag{22}$$

## VII. Assignment Constraints

### A. Single Assignment per Survivor

Each survivor in $\mathcal{Q}_{\text{high}}(t)$ appears in at most one tour:

$$\sum_j \mathbf{1}\{i \in \pi_j(t)\} \le 1 \quad \forall i \in \mathcal{Q}_{\text{high}}(t) \tag{23}$$

### B. Capacity per Tour

Each tour respects item capacity:

$$\sum_{i\in\pi_j(t)} |\mathcal{R}_i| \le K \quad \forall j \in \mathcal{D}_{\text{avail}}(t) \tag{24}$$

where $K$ is the item capacity (e.g., 4), $\text{len}(\pi_j) = k$ is the number of survivors and $\sum |\mathcal{R}_i|$ is the number of items. A UAV may stay idle ($\pi_j = \emptyset$) if no feasible tour exists.

### C. Feasibility Enforcement

If a tour $\pi_j$ is not feasible, its cost is set to infinity: $C(\pi_j) = \infty$ when $\neg\text{Feasible}(\pi_j)$ (includes $c_i < c_{\text{thresh}}$, $A_j = 0$, $P_l \not\subset V$, $\mathcal{R}_{i_l} \not\subseteq I_j^{(l-1)}$, or $E(\pi_j) + E_{\text{reserve}} > B_j$). Any optimal solution will have such tours excluded.

## VIII. Objective Function

### A. Primary Objective: Minimize Cost

The snapshot problem OFF-SNAP$(t)$ over tours $\{\pi_j\}$ is:

$$\min_{\{\pi_j\}} \sum_{j\in\mathcal{D}_{\text{avail}}(t)} C(\pi_j) \tag{25}$$

where

$$C(\pi_j) = \sum_{i\in\pi_j} \left[ w_T \frac{T_{\text{leg}}(i)}{T_{\max}} + w_E \frac{E_{\text{leg}}(i)}{E_{\max}} - w_U u_i(t) c_i \right] \tag{26}$$

Note that the single-assignment cost $C_{ij}(t)$ in §5.6 is the $k = 1$ special case $C(\pi_j = (i)) = C_{ij}(t)$. where $T_{\text{leg}}(i), E_{\text{leg}}(i)$ include travel to $i$ plus 15s hover. If $\mathcal{Q}_{\text{high}} = \emptyset$ or all $C(\pi_j) = \infty$, the optimum is $\pi_j = \emptyset$ (idle).

### B. State Update for Tours

Dispatching $\pi_j = (i_1, \dots, i_k)$ updates: $\mathcal{Q}(t^+) = \mathcal{Q}(t^-) \setminus \{i_1, \dots, i_k\}$, $B_j(t^+) = B_j(t^-) - E(\pi_j)$, $I_j(t^+) = I_j(t^-) \setminus \bigcup_{i\in\pi_j} \mathcal{R}_i$, $m(t^+) = m(t^-) - \sum_{i\in\pi_j} weight(\mathcal{R}_i)$, $\text{status}_j \leftarrow$ EnRoute until return to $p_j^{\text{base}}$ for reload.

## IX. State Transition Functions

### A. New Detection

$$\mathcal{S}(t^+) = \mathcal{S}(t^-) \cup \{S_{\text{new}}\} \tag{27}$$

$$\mathcal{Q}(t^+) = \mathcal{Q}(t^-) \cup \{S_{\text{new}}\} \tag{28}$$

### B. Tour Dispatch

Dispatching tour $\pi_j = (i_1, \dots, i_k)$:

$$\mathcal{Q}(t^+) = \mathcal{Q}(t^-) \setminus \{i_1, \dots, i_k\} \tag{29}$$

$$B_j(t^+) = B_j(t^-) - E(\pi_j) \tag{30}$$

$$I_j(t^+) = I_j(t^-) \setminus \bigcup_{i\in\pi_j} \mathcal{R}_i \tag{31}$$

$$m(t^+) = \sum_{item\in I_j(t^+)} weight(item) \tag{32}$$

$$\text{status}_j \leftarrow \text{EnRoute} \tag{33}$$

### C. Tour Completes (Return to Base)

Upon return to $p_j^{\text{base}}$:

$$\text{status}_j \leftarrow \text{Idle} \tag{34}$$

$$p_j \leftarrow p_j^{\text{base}} \tag{35}$$

$$I_j \leftarrow \text{reload}(K) \tag{36}$$

$$m \leftarrow \sum_{item\in I_j} weight(item) \tag{37}$$

$$A_j \leftarrow 1 \tag{38}$$

### D. Expiration and Fairness

TABLE IV
Expiry/Fairness Parameters

| **Parameter** | **Value** | **Meaning** |
|---|---|---|
| $T_{\text{exp}}$ | 90 min | Max wait before expiry (remove from $\mathcal{Q}$) |
| $u_{\min}$ | 0.05 | Min urgency threshold ($u < u_{\min} \Rightarrow$ expire) |
| $T_{\text{fair}}$ | 15 min | Fairness wait for $\sigma_i > 0.8$ (promote to top of $\mathcal{Q}_{\text{high}}$) |

Expire: if $t - t_i^{\text{det}} > T_{\text{exp}}$ or $u_i(t) < u_{\min} \Rightarrow \mathcal{Q} \leftarrow \mathcal{Q} \setminus \{S_i\}$. Fairness: if $\sigma_i > 0.8 \wedge (t - t_i^{\text{det}}) > T_{\text{fair}} \Rightarrow$ promote $S_i$ to top of $\mathcal{Q}_{\text{high}}$.

## X. Scope and Limitations

### A. Core Model Includes

- Survivor detection as exogenous input
- Responder UAV assignment and routing
- Battery-aware feasibility
- Time-dependent urgency
- Online, event-triggered decision making

### B. Excluded (Future Work)

- Dynamic aftershock obstacles
- UAV-to-UAV collision avoidance
- Communication relay UAVs
- Hot-swap battery stations
- Detection uncertainty verification policies
- Recourse/reassignment after dispatch
- High-fidelity nonlinear energy/power model (mass, aerodynamic drag, wind, acceleration-aware)

## XI. Conclusion

This document presents a complete mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM) for post-earthquake UAV-assisted response. The model captures the essential dynamics of the system: survivor states with time-dependent urgency, Responder UAV states with battery constraints, and the online nature of detection arrivals and resource availability. The formulation separates feasibility (battery and routing constraints) from optimization (assignment and path selection), providing a clear foundation for algorithm design and simulation. The coupling between assignment and routing is explicitly defined through the tour cost function $C(\pi_j)$ (with $C_{ij} = C(\pi_j = (i))$ as the $k = 1$ special case), which depends on travel time and energy per leg and urgency per survivor.